

WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.

1	2	3	4	5	6	7	8						
A								A					
B								B					
C								C					
D								D					
E								E					
F								F					
				Date Prep.	15.11.2018	PROJECT TITLE		ABB	EPMV	COVER SHEET MVD-22 20kV Central WTP Area SCHEMATIC DIAGRAM SBC-W OUTGOING	Scale	NTS	= H02,H03
				Prepared	YR	MV Switchgear for MNA Serang Plant					Customer	PT Multimas Nabati Asahan	& 306
00	Issued for Approval	15.01.19	YR	Checked	AW			PT. ABB Sakti Industri	Drawing Nbr.		18.042-MNA-306.04	Sh.	1
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042			Customer Doc. Nbr.			Cont.	27
1	2	3	4	5	6	7	8						

UniSec

Medium Voltage Switchgear
PT MULTIMAS NABATI ASAHAN
MVD-22 20kV Central WTP Area
SCHEMATIC DIAGRAM
SBC-W OUTGOING
H02,H03

SHEET INDEX

PAGE	PAGE NAME	REVISION		
==MVD-22=H02,H03&306/1	COVER SHEET	00		
==MVD-22=H02,H03&306/2	Index of Sheet	00		
==MVD-22=H02,H03&306/3	REFERENCE DESIGNATIONS	00		
==MVD-22=H02,H03&306/4	REFERENCE DESIGNATIONS	00		
==MVD-22=H02,H03&306/5	REFERENCE DESIGNATIONS	00		
==MVD-22=H02,H03&306/6	REFERENCE DESIGNATIONS	00		
==MVD-22=H02,H03&306/7	DISTRIBUTION OF AUXILIARY CIRCUITS	00		
==MVD-22=H02,H03&306/8	AUXILIARY SUPPLY CIRCUIT	00		
==MVD-22=H02,H03&306/9	CIRCUIT BREAKER CONTROL	00		
==MVD-22=H02,H03&306/10	AUXILIARY MOTOR & RELAY	00		
==MVD-22=H02,H03&306/11	DIGITAL INPUT/OUTPUT RELAY	00		
==MVD-22=H02,H03&306/12	SPARE CONTACT	00		
==MVD-22=H02,H03&306/13	CURRENT TRANSFORMER	00		
==MVD-22=H02,H03&306/14	VOLTAGE TRANSFORMER	00		
==MVD-22=H02,H03&306/15	LV DOOR LAYOUT	00		
==MVD-22=H02,H03&306/16	LV BOX LAYOUT	00		
==MVD-22=H02,H03&306/17	PART LIST	00		
==MVD-22=H02,H03&306/18	Terminal diagram ==MVD-22=H02,H03+-XDA			
==MVD-22=H02,H03&306/19	Terminal diagram ==MVD-22=H02,H03+-XDB1			
==MVD-22=H02,H03&306/20	Terminal diagram ==MVD-22=H02,H03+-XDC1			
==MVD-22=H02,H03&306/21	Terminal diagram ==MVD-22=H02,H03+-XDC2			
==MVD-22=H02,H03&306/22	Terminal diagram ==MVD-22=H02,H03+-XDI			
==MVD-22=H02,H03&306/23	Terminal diagram ==MVD-22=H02,H03+-XDI2			
==MVD-22=H02,H03&306/24	Terminal diagram ==MVD-22=H02,H03+-XDI3			
==MVD-22=H02,H03&306/25	Terminal diagram ==MVD-22=H02,H03+-XDI6			
==MVD-22=H02,H03&306/26	Terminal diagram ==MVD-22=H02,H03+-XDV			
==MVD-22=H02,H03&306/27	Terminal diagram ==MVD-22=H02,H03+-XDX			

[illegible]

WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.

A	1				2				3				4				5				6				7				8											
	REFERENCE DESIGNATION OF OBJECTS IN ELECTRICAL DOCUMENTS																				-BGB5 POSITION SWITCH OF CIRCUIT-BREAKER SIGNALLING UNDERVOLTAGE RELEASE ENERGIZED																			
B	(IN COMPLIANCE WITH STANDARD IEC 81346-2 AND ABB TECHNICAL STANDARD 2NBA000001)																				-BGB6 POSITION SWITCH OF CIRCUIT-BREAKER SIGNALLING UNDERVOLTAGE RELEASE EXCLUDED MECHANICALLY																			
	DESIGNATION																				DESCRIPTION																			
C	-AA MULTIFUNCTION UNIT (CENTRAL UNIT)																				-BGB7 POSITION SWITCHES OF CIRCUIT-BREAKER ON SWITCHGEAR																			
	-BAD1,-BAD4 CAPACITIVE VOLTAGE DIVIDER LOCATED ON PHASE L1																				-BGB8...-BGB10 POSITION SWITCHES OF CIRCUIT-BREAKER																			
	-BAD2,-BAD5 CAPACITIVE VOLTAGE DIVIDER LOCATED ON PHASE L2																				-BGB11 POSITION SWITCH OF CIRCUIT-BREAKER OPERATING FOR MANUAL OPENING																			
	-BAD3,-BAD6 CAPACITIVE VOLTAGE DIVIDER LOCATED ON PHASE L3																				-BGB12 PROXIMITY SWITCH SIGNALLING CIRCUIT-BREAKER IN OPEN POSITION																			
	-BAR VOLTAGE PROTECTION RELAY																				-BGB13 PROXIMITY SWITCH SIGNALLING CIRCUIT-BREAKER IN CLOSED POSITION																			
	-BAS1 VOLTAGE SENSOR LOCATED ON PHASE L1																				-BGD POSITION SWITCH OF L.V. INSTRUMENT COMPARTMENT DOOR.																			
	-BAS2 VOLTAGE SENSOR LOCATED ON PHASE L2																				-BGD1 POSITION SWITCH OF CIRCUIT-BREAKER COMPARTMENT DOOR																			
	-BAS3 VOLTAGE SENSOR LOCATED ON PHASE L3																				-BGD2 POSITION SWITCH OF CABLE COMPARTMENT DOOR																			
	-BAT1 VOLTAGE TRANSFORMER LOCATED ON PHASE L1																				-BGE1 POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE IN OPEN POSITION																			
	-BAT2 VOLTAGE TRANSFORMER LOCATED ON PHASE L2																				-BGE2 POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE IN CLOSED POSITION																			
D	-BAT3 VOLTAGE TRANSFORMER LOCATED ON PHASE L3																				-BGE3 POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)																			
	-BAT4 VOLTAGE TRANSFORMER IN BUSBAR COMPARTMENT LOCATED ON PHASE L1																				-BGE4 POSITION SWITCHES OF OPERATION OF EARTHING SWITCH -QCE ACTUATED BY ELECTRIC MOTOR																			
	-BAT5 VOLTAGE TRANSFORMER IN BUSBAR COMPARTMENT LOCATED ON PHASE L2																				-BGE5 POSITION SWITCHES OF OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2 ACTUATED BY ELECTRIC MOTOR																			
	-BAT6 VOLTAGE TRANSFORMER IN BUSBAR COMPARTMENT LOCATED ON PHASE L3																				-BGE7 POSITION SWITCH SIGNALLING EARTHING SWITCH NOT IN MANUAL OPERATION																			
	-BCD DIFFERENTIAL PROTECTIVE RELAY																				-BGE8 POSITION SWITCHES OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH- DISCONNECTOR -QBS																			
	-BCF FEEDER PROTECTION RELAY																				-BGE11 POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE1 OR -QCE2 IN OPEN POSITION																			
	-BCG GENERATOR PROTECTION RELAY																				-BGE12 POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE1 OR -QCE2 IN CLOSED POSITION																			
	-BCM MOTOR PROTECTION RELAY																				-BGE13 POSITION SWITCHES SIGNALLING EARTHING SWITCH -QCE1 OR -QCE2 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)																			
	-BCN NEUTRAL (RESIDUAL) CURRENT TRANSFORMER																				-BGE14, -BGE15 POSITION SWITCHES FOR OPERATION OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH-DISCONNECTOR -QBS, ACTUATED BY ELECTRIC MOTOR																			
	-BCP TRANSFORMER PROTECTION RELAY																				-BGF POSITION SWITCHES OF MEDIUM VOLTAGE FUSES																			
E	-BCR CURRENT PROTECTION RELAY																				-BGI1 POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD IN OPEN POSITION																			
	-BCS1 CURRENT SENSOR LOCATED ON PHASE L1																				-BGI2 POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD IN CLOSED POSITION																			
	-BCS2 CURRENT SENSOR LOCATED ON PHASE L2																				-BGI3 POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)																			
	-BCS3 CURRENT SENSOR LOCATED ON PHASE L3																				-BGI4 POSITION SWITCHES OF OPERATION OF DISCONNECTOR -QBD1 ACTUATED BY ELECTRIC MOTOR																			
	-BCT1 CURRENT TRANSFORMER LOCATED ON PHASE L1																				-BGI5 POSITION SWITCHES OF OPERATION OF DISCONNECTOR -QBD2 ACTUATED BY ELECTRIC MOTOR																			
	-BCT2 CURRENT TRANSFORMER LOCATED ON PHASE L2																				-BGI6 POSITION SWITCH SIGNALLING DISCONNECTOR (OR SWITCH-DISCONNECTOR) -QBD NOT IN EMERGENCY MANUAL OPERATION																			
	-BCT3 CURRENT TRANSFORMER LOCATED ON PHASE L3																				-BGI7 POSITION SWITCH SIGNALLING DICONNECTOR (OR SWITCH-DISCONNECTOR) -QBD NOT IN MANUAL OPERATION																			
	-BCT4,-BCT7 CURRENT TRANSFORMER LOCATED ON PHASE L1																				-BGI8 POSITION SWITCHES OF SWITCH-DISCONNECTOR -QBS																			
	-BCT5,-BCT8 CURRENT TRANSFORMER LOCATED ON PHASE L2																				-BGI9 POSITION SWITCHES OF OPERATION OF SWITCH-DISCONNECTOR -QBS ACTUATED BY ELECTRIC MOTOR																			
	-BCT6,-BCT9 CURRENT TRANSFORMER LOCATED ON PHASE L3																				-BGI11 POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD1 IN OPEN POSITION																			
F	-BCZ DISTANCE PROTECTION RELAY																				-BGI12 POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD1 IN CLOSED POSITION																			
	-BEF FREQUENCY PROTECTION RELAY																				-BGI13 POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD1 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)																			
	-BER SUPERVISION RELAYS																				-BGI14, -BGI15 POSITION SWITCHES OF OPERATION OF DISCONNECTOR -QBD1 ACTUATED BY ELECTRIC MOTOR																			
	-BES SYNCHRONIZING RELAY																				-BGI21 POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD2 IN OPEN POSITION																			
	-BET THERMAL PROTECTION RELAY																				-BGI22 POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD2 IN CLOSED POSITION																			
	-BGA1 POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN CIRCUIT-BREAKER COMPARTMENT																				-BGI23 POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD2 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)																			
	-BGA2 POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN BUSBAR COMPARTMENT (SINGLE BUSBAR SYSTEM) OR IN FRONT BUSBAR COMPARTMENT (DOUBLE BUSBAR SYSTEM)																				-BGI24, -BGI25 POSITION SWITCHES OF OPERATION OF DISCONNECTOR -QBD2 ACTUATED BY ELECTRIC MOTOR																			
	-BGA3 POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN CABLE COMPARTMENT																																							
	-BGA4 POSITION SWITCH FOR DETECTION OF INTERNAL ARCING IN REAR BUSBAR COMPARTMENT (DOUBLE BUSBAR SYSTEM)																																							
	-BGB1...-BGB3 POSITION SWITCHES OF CIRCUIT-BREAKER																																							
-BGB4 POSITION SWITCH OF CIRCUIT-BREAKER: PASSING CONTACT CLOSING MOMENTARILY WHEN THE BREAKER OPENS																																								
				Date Prep. 15.11.2018				PROJECT TITLE				ABB EPMV				REFERENCE DESIGNATIONS				Scale				NTS				= H02,H03												
				Prepared YR				MV Switchgear for MNA Serang Plant								MVD-22 20kV Central WTP Area				Customer				PT Multimas Nabati Asahan				& 306												
00				Issued for Approval 15.01.19 YR				Checked AW								SCHEMATIC DIAGRAM				Drawing Nbr.				18.042-MNA-306.04				Sh. 3												
Ind.				Revision				Date				Name				Approved AI				Based on: BE318042				PT. ABB Sakti Industri				SBC-W OUTGOING				Customer Doc. Nbr.				Cont. 27				
1				2				3				4				5				6				7				8												

WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.

A	-BGI31				POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION				-CA				CAPACITORS				A														
	-BGI32				POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN CLOSED POSITION				-CA1				CAPACITOR FOR FRONT VENTILATOR																		
	-BGI33				POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD3 IN OPEN POSITION AND NOT IN OPERATION (OPERATING LEVER NOT INSERTED)				-CA2				CAPACITOR FOR REAR VENTILATOR																		
	-BGI41...				POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN FEEDER POSITION				-EA1				LIGHTING LAMP LOCATED IN L.V. INSTRUMENT COMPARTMENT.																		
	-BGI42				POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS IN OPEN POSITION				-EA2				LIGHTING LAMP LOCATED IN CABLE COMPARTMENT																		
B	-BGI43...				POSITION SWITCHES SIGNALLING SWITCH-DISCONNECTOR -QBS CLOSED IN EARTH POSITION				-EA3				LIGHTING LAMP LOCATED IN OPERATING MECHANISM ENCLOSURE				B														
	-BGI51				POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN FEEDER POSITION				-EA4				LIGHTING LAMP LOCATED IN CIRCUIT BREAKER COMPARTIMENT																		
	-BGI52				POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 IN OPEN POSITION				-EB1				HEATER LOCATED IN CABLE COMPARTMENT																		
	-BGI53				POSITION SWITCHES SIGNALLING DISCONNECTOR -QBD5 CLOSED IN EARTH POSITION				-EB2				HEATER LOCATED IN CIRCUIT-BREAKER COMPARTMENT																		
	-BGK				POSITION SWITCH OF KEY LOCK				-EB3				HEATER LOCATED IN MOTOR CABINET																		
C	-BGL1				POSITION SWITCH OF ELECTROMECHANICAL LOCK -RLE1				-EB4				HEATER FOR VENTILATION TEST				C														
	-BGS1...-BGS4				POSITION SWITCHES OF CIRCUIT-BREAKER SPRINGS				-EB5				HEATER LOCATED IN L.V. INSTRUMET COMPARTMENT.																		
	-BGS5				PROXIMITY SWITCH SIGNALLING SPRINGS IN CHARGED POSITION				-EB7				HEATERS LOCATED IN LEFT SIDE OF THE OPERATING MECHANISM ENCLOSURE																		
	-BGT1				POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN SERVICE POSITION				-EB8				HEATERS LOCATED IN RIGHT SIDE OF THE OPERATING MECHANISM ENCLOSURE																		
	-BGT2				POSITION SWITCHES ON TRUCK SIGNALLING TRUCK IN TEST POSITION				-FA1				SURGE ARRESTER LOCATED ON PHASE L1																		
D	-BGT3				POSITION SWITCH ON TRUCK SIGNALLING TRUCK NOT IN ISOLATING TRAVEL POSITION				-FA2				SURGE ARRESTER LOCATED ON PHASE L2				D														
	-BGT4				POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN SERVICE POSITION				-FA3				SURGE ARRESTER LOCATED ON PHASE L3																		
	-BGT5				POSITION SWITCHES ON SWITCHGEAR SIGNALLING TRUCK IN TEST POSITION				-FCD				FUSE-DISCONNECTORS FOR PROTECTION OF AUXILIARY CIRCUITS																		
	-BGT6				POSITION SWITCHES SIGNALLING V.T. TRUCK IN SERVICE POSITION				-FCF1				MEDIUM VOLTAGE FUSE LOCATED ON PHASE L1																		
	-BGT7				POSITION SWITCHES SIGNALLING V.T. TRUCK IN TEST POSITION				-FCF2				MEDIUM VOLTAGE FUSE LOCATED ON PHASE L2																		
E	-BGT8				POSITION SWITCHES SIGNALLING EARTHING TRUCK IN SERVICE POSITION				-FCF3				MEDIUM VOLTAGE FUSE LOCATED ON PHASE L3				E														
	-BGT11				PROXIMITY SWITCH SIGNALLING TRUCK IN SERVICE POSITION				-FCM1				MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF SPRINGS CHARGING MOTOR ON MAIN CIRCUIT-BREAKER																		
	-BGT12				PROXIMITY SWITCH SIGNALLING TRUCK IN TEST POSITION				-FCM2...-FCM10				MINIATURE CIRCUIT-BREAKERS FOR PROTECTION OF AUXILIARY CIRCUITS																		
	-BGV1				POSITION SWITCH OF FRONT VENTILATOR				-FCM11				MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV1																		
	-BGV2				POSITION SWITCH OF REAR VENTILATOR				-FCM12				MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV2																		
F	-BM				HYGROSTAT				-FCM13				MINIATURE CIRCUIT-BREAKER FOR PROTECTION OF VENTILATOR -MV3				F														
	-BPA4				PRESSURE SWITCH LOCATED IN CABLE COMPARTMENT FOR DETECTION OF INTERNAL ARCING				-FCT				THERMAL OVERLOAD RELEASES																		
	-BPA5				PRESSURE SWITCH LOCATED IN CIRCUIT-BREAKER OR IN UPPER VT COMPARTMENT FOR DETECTION OF INTERNAL ARCING				-GA				GENERATORS																		
	-BPA6				PRESSURE SWITCH LOCATED IN BUSBAR OR IN BUSBAR AND CIRCUIT-BREAKER COMPARTMENT FOR DETECTION OF INTERNAL ARCING				-GQ1				FRONT VENTILATOR																		
	-BPA7				PRESSURE SWITCH LOCATED IN VOLTAGE TRANSFORMER ISOLATING COMPARTMENT FOR DETECTION OF INTERNAL ARCING				-GQ2				REAR VENTILATOR																		
-BPS1				PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER OR ON POLE OF PHASE L1 OF CIRCUIT-BREAKER				-GQ3				VENTILATOR LOCATED ON CIRCUIT-BREAKER TRUCK																			
-BPS2				PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L2				-KFA1				AUXILIARY RELAY SIGNALLING LOW GAS PRESSURE																			
-BPS3				PRESSURE SWITCH LOCATED ON CIRCUIT-BREAKER POLE OF PHASE L3				-KFA2				AUXILIARY RELAY SIGNALLING INSUFFICIENT GAS PRESSURE																			
-BR				FLAME DETECTORS, SMOKE DETECTORS				-KFA3...-KFA9				AUXILIARY RELAYS OR CONTACTORS																			
-BT				THERMOSTAT				-KFC				CLOSING RELAYS OR CONTACTORS																			
-BT1				THERMOSTAT FOR DE-ENERGIZATION OF VENTILATORS				-KFI				INTEGRATED CIRCUITS																			
-BT2				THERMOSTAT FOR ENERGIZATION OF VENTILATORS				-KFL				LOCKOUT RELAY																			
-BT3				THERMOSTAT SIGNALLING HIGH TEMPERATURE ALARM				-KFM				MICROPROCESSORS																			
-BU				BUCHHOLZ RELAY				-KFN				ANTIPUMPING RELAYS																			
-BX1				UNIT WITH SENSORS FOR DETECTION OF INTERNAL ARCING				-KFO				OPENING RELAYS OR CONTACTORS																			
-BX2				ADDITIONAL CURRENT SENSING UNIT FOR DETECTION OF INTERNAL ARCING				-KFP				PROGRAMMABLE LOGIC CONTROLLERS (PLC)																			
-BZ,-BZ4				MULTIFUNCTION SENSORS				-KFR				RECLOSING RELAYS																			
-BZ1				COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L1				-KFS				SYNCHRONIZING DEVICES																			
-BZ2				COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L2				-KFT				AUXILIARY TIME RELAYS, DELAY ELEMENTS																			
-BZ3				COMBINED CURRENT AND VOLTAGE SENSOR, LOCATED ON PHASE L3				-KFZ				CONTROLLERS																			
								-MBC				CLOSING RELEASE OF CIRCUIT-BREAKER																			
								-MAD				MOTOR FOR ELECTRICAL OPERATION OF DISCONNECTOR -QBD																			
								-MAD1				MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD1																			
								-MAD2				MOTOR FOR ELECTRICAL OPERATION OF DICONNECTOR -QBD2																			
								-MAD9				MOTOR FOR ELECTRICAL OPERATION OF SWITCH-DISCONNECTOR -QBS																			
				Date Prep.		15.11.2018		PROJECT TITLE MV Switchgear for MNA Serang Plant		ABB EPMV		REFERENCE DESIGNATIONS MVD-22 20kV Central WTP Area SCHEMATIC DIAGRAM SBC-W OUTGOING		Scale		NTS		= H02,H03													
				Prepared		YR								Customer		PT Multimas Nabati Asahan		& 306													
00		Issued for Approval		15.01.19		YR								Checked		AW		Drawing Nbr.		18.042-MNA-306.04		Sh. 4									
Ind.		Revision		Date		Name		Approved		AI		Based on:		BE318042		PT. ABB Sakti Industri		Customer Doc. Nbr.				Cont. 27									
1				2				3				4				5				6				7				8			

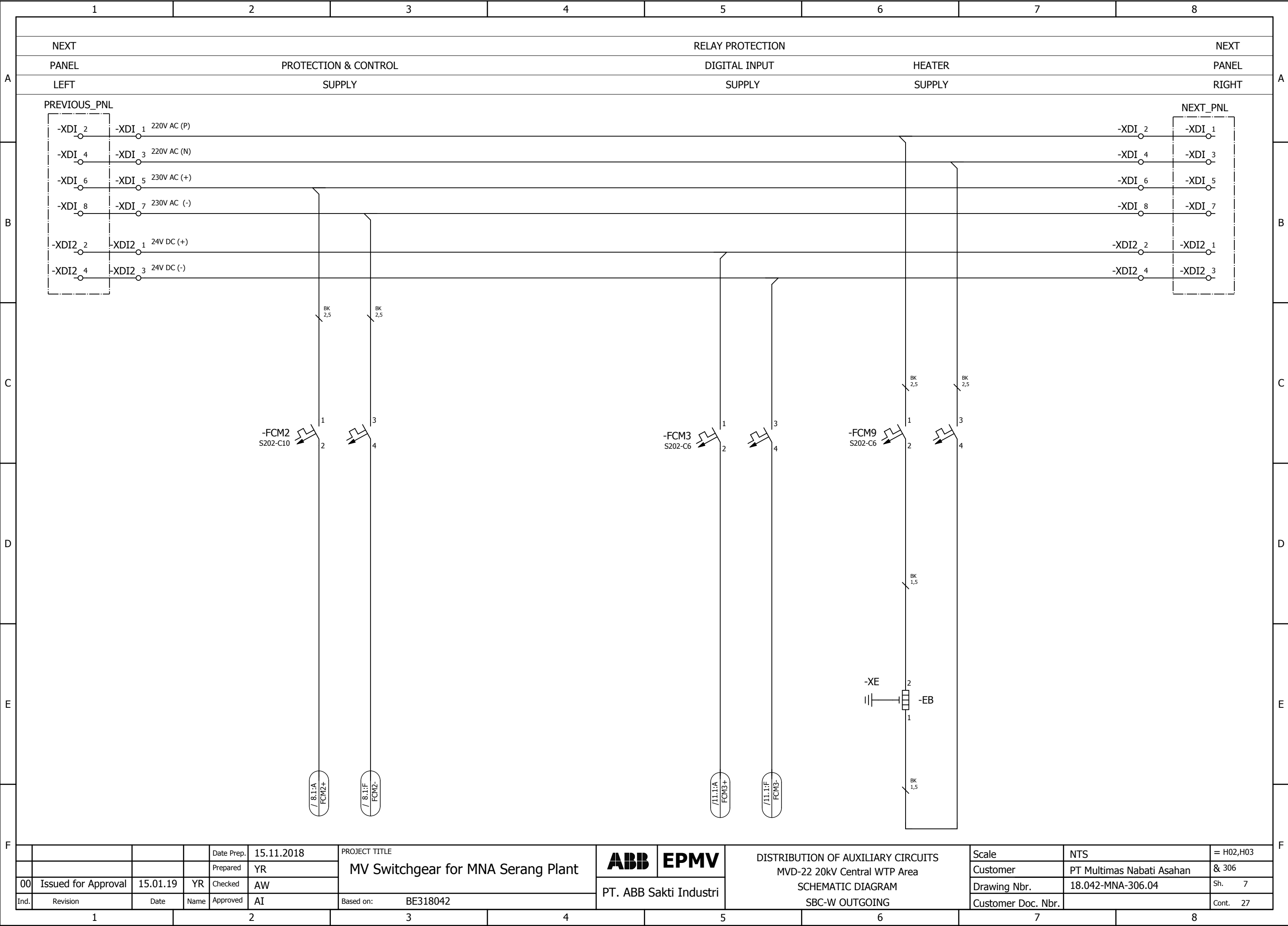
WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.

1		2		3		4		5		6		7		8				
A	-MAE	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE						-QBH	MANUAL CIRCUIT-BREAKERS							A		
	-MAE1	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2						-QBL	LINKS									
	-MAE8	MOTOR FOR ELECTRICAL OPERATION OF EARTHING SWITCH -QCE SUPPLIED TOGETHER WITH SWITCH-DISCONNECTOR -QBS						-QBM	MINIATURE SWITCH-DISCONNECTORS									
	-MAS	MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING						-QBS	SWITCH-DISCONNECTORS									
	-MAT	MOTOR FOR ELECTRICAL OPERATION OF TRUCK RACKING-IN/OUT						-QBT	TRUCK									
	-MBC	CLOSING RELEASE OF CIRCUIT-BREAKER						-QCA	EARTHING SWITCH FOR ELIMINATION OF INTERNAL ARCING									
	-MBC4	CLOSING RELEASE OF SWITCH-DISCONNECTOR -QBS						-QCE	EARTHING SWITCH									
	-MBO1	FIRST OPENING RELEASE OF CIRCUIT-BREAKER						-QCE1	EARTHING SWITCH OF BUSBARS (SINGLE BUSBAR SYSTEM) OR OF FRONT BUSBARS (DOUBLE BUSBAR SYSTEM)									
	-MBO2	SECOND OPENING RELEASE OF CIRCUIT-BREAKER						-QCE2	EARTHING SWITCH OF REAR BUSBARS (DOUBLE BUSBAR SYSTEM)									
	-MBO3	OPENING SOLENOID FOR OVERCURRENT RELEASE OF CIRCUIT-BREAKER OR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER						-QQ	CLUTCHES									
	B	-MBO4	OPENING RELEASE OF SWITCH-DISCONNECTOR -QBS						-RAA	ANTIFERRORESONANCE RESISTOR							B	
		-MBU	UNDERVOLTAGE RELEASE OF CIRCUIT-BREAKER						-RAD	DIODES								
-PFB		BLUE SIGNAL LAMPS						-RAI	INDUCTORS									
-PFF		FLAG RELAYS						-RAR	RESISTORS									
	-PFG	GREEN SIGNAL LAMPS						-RF	FILTERS									
	-PFR	RED SIGNAL LAMPS						-RLE1	ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING									
	-PFS	SHORT CIRCUIT INDICATORS						-RLE2	ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT									
	-PFV	VOLTAGE INDICATORS						-RLE3	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE									
	-PFV1	VOLTAGE INDICATORS ON FEEDER SIDE						-RLE4	ELECTROMECHANICAL LOCK PREVENTING THE DOOR OPENING OPERATION									
	-PFV2	VOLTAGE INDICATORS ON BUSBAR SIDE						-RLE5	ELECTROMECHANICAL LOCK PREVENTING OPERATION OF DISCONNECTOR -QBD									
	C	-PFW	WHITE SIGNAL LAMPS						-RLE6	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD1							C	
		-PFX	CROSS INDICATORS, ELECTROMECHANICAL INDICATORS						-RLE7	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF DISCONNECTOR -QBD2								
-PFY		YELLOW SIGNAL LAMPS						-RLE8	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR CLOSING OPERATION OF EARTHING SWITCH -QCE1 OR -QCE2									
-PGA		AMMETERS						-RLE9	ELECTROMECHANICAL LOCK PREVENTING OPERATION OF SWITCH-DISCONNECTOR -QBS									
	-PGC	COUNTERS						-RLE11	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD4									
	-PGF	FREQUENCYMETERS						-RLE12	ELECTROMECHANICAL LOCK PREVENTING INSERTION OF LEVER FOR OPERATION OF DISCONNECTOR -QBD5									
	-PGH	HOURMETERS						-SFA	AMMETRIC SWITCHES									
	-PGI	PROTECTION AND CONTROL UNIT: HUMAN MACHINE INTERFACE						-SFC	CONTROL SWITCHES, CLOSING PUSH-BUTTONS									
	-PGI1	HUMAN MACHINE INTERFACE OF ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON CIRCUIT-BREAKER)						-SFL	LOCKING CONTACTS									
	-PGI2	HUMAN MACHINE INTERFACE FOR ELECTRONIC CONTROLLED CIRCUIT-BREAKER (ON SWITCHGEAR)						-SFO	OPENING PUSH-BUTTONS									
	-PGJ	ACTIVE ENERGY METERS						-SFR	RESET PUSH-BUTTONS									
	-PGK	REACTIVE ENERGY METERS						-SFS	SELECTOR SWITCHES									
	-PGM	MULTIFUNCTION INDICATORS						-SFT	TEST PUSH-BUTTONS									
	-PGP	POWER-FACTOR METERS						-SFU	UNLOCKING PUSH-BUTTONS									
	-PGQ	VARMETERS						-SFV	VOLTMETRIC SWITCHES									
	-PGS	SYNCHRONOSCOPES						-TA	POWER TRANSFORMERS									
	-PGV	VOLTMETERS						-TB	CONVERTER									
	-PGW	WATTMETERS						-TB1	RECTIFIER FOR FIRST OPENING RELEASE									
	-PJ	ACOUSTICAL SIGNAL DEVICES (BELLS, SIRENS)						-TB2	RECTIFIER FOR SECOND OPENING RELEASE									
	-QAB	CIRCUIT-BREAKERS						-TB3	RECTIFIER FOR CLOSING RELEASE									
	-QAC	CONTACTORS (FOR POWER)						-TB4	RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING CIRCUIT-BREAKER CLOSING									
	-QBD	DISCONNECTORS						-TB5	RECTIFIER FOR ELECTROMECHANICAL LOCK PREVENTING TRUCK RACKING-IN/OUT									
	-QBD1	FRONT BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)						-TB6	RECTIFIER FOR UNDERVOLTAGE RELEASE									
	-QBD2	REAR BUSBAR DISCONNECTORS (DOUBLE BUSBAR SYSTEM)						-TB7	ERECTIFIER FOR INDIRECT OVERCURRENT RELEASE ON CIRCUIT-BREAKER									
	-QBD3	FEEDER DISCONNECTORS (DOUBLE BUSBAR SYSTEM)														F		
	-QBD4	TWO-WAY DISCONNECTOR																
	-QBD5	TWO-WAY DISCONNECTOR ON BUSBAR RISER																
1		2		3		4		5		6		7		8				
				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant		ABB	EPMV	REFERENCE DESIGNATIONS MVD-22 20kV Central WTP Area		Scale	NTS		= H02,H03			
				Prepared	YR							Customer	PT Multimas Nabati Asahan		& 306			
00	Issued for Approval	15.01.19	YR	Checked	AW			PT. ABB Sakti Industri		SCHEMATIC DIAGRAM SBC-W OUTGOING		Drawing Nbr.	18.042-MNA-306.04		Sh. 5			
Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318042					Customer Doc. Nbr.			Cont. 27			

WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.

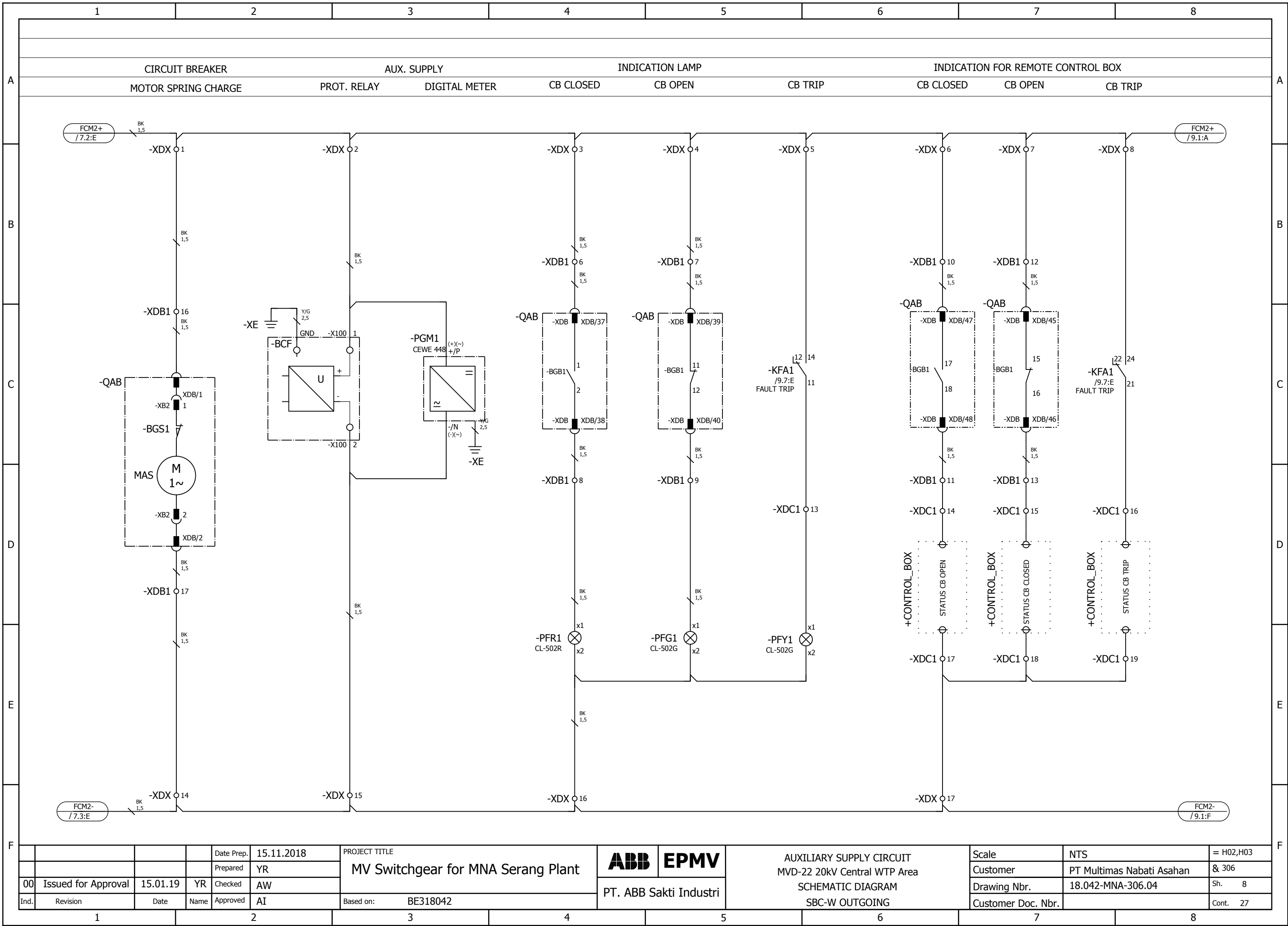
A	-TFA				ACTIVE POWER TRANSDUCERS				-XDI1	TERMINAL BLOCK FOR INTERCONNECTIONS OF CONTROL CIRCUITS				A										
	-TFC				CURRENT TRANSDUCERS					-XDI2	TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR CIRCUIT-BREAKER SPRINGS CHARGING													
	-TFF				FREQUENCY TRANSDUCERS				-XDI3		TERMINAL BLOCK FOR INTERCONNECTIONS OF CIRCUITS OF MOTOR FOR SWITCH-DISCONNECTOR OPERATION													
	-TFJ				ACTIVE ENERGY TRANSDUCERS					-XDI4	TERMINAL BLOCK FOR INTERCONNECTIONS OF A.C. AUXILIARY CIRCUITS													
	-TFK				REACTIVE ENERGY TRANSDUCERS				-XDI6		TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS) OF VOLTAGE CIRCUITS													
	-TFM				MULTIFUNCTION TRANSDUCERS					-XDI7	TERMINAL BLOCK FOR INTERCONNECTION OF MOD-BUS													
	-TFP				POWER-FACTOR TRANSDUCERS				-XDI8		TERMINAL BLOCK FOR INTERCONNECTIONS OF CURRENT CIRCUITS													
	-TFQ				REACTIVE POWER TRANSDUCERS					-XDI9	TERMINAL BLOCK FOR SPECIAL USE													
	-TFS				SIGNAL CONVERTERS				-XDJ		TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF DISCONNECTORS -QBD													
	-TFV				VOLTAGE TRANSDUCERS					-XDJ1	TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR -QBS													
B	-WA				BUSBARS				-XDM		SEALABLE TERMINAL BLOCK FOR MEASUREMENT				B									
	-WBC				POWER CABLES					-XDS	CURRENT SOCKETS													
	-WE				EARTHING CABLES FOR L.V. INSTRUMENT COMPARTMENT.				-XDT		TERMINAL BLOCK FOR POSITION CONTACTS OF TRUCK AND OF CONNECTOR -XDB													
	-WGC				CONTROL CABLES					-XDU	CONNECTOR FOR ISOLATION OF CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK													
	-WGS				COMMUNICATION CABLES (SHIELDED TWISTED PAIRS)				-XDU1		TERMINAL BLOCK FOR CIRCUITS OF LOWER VOLTAGE TRANSFORMER TRUCK													
	-WH				OPTICAL FIBRES					-XDV	TERMINAL BLOCK FOR CIRCUITS OF VOLTAGE TRANSFORMERS													
	-XDA				TERMINAL BLOCK FOR CIRCUITS OF CURRENT TRANSFORMERS				-XDV1		CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L1													
	-XDA1				CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L1					-XDV2	CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L2													
	-XDA2				CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L2				-XDV3		CONNECTOR FOR CIRCUITS OF VOLTAGE TRANSFORMERS LOCATED ON PHASE L3													
	-XDA3				CONNECTOR FOR CIRCUITS OF CURRENT TRANSFORMERS LOCATED ON PHASE L3					-XDV4	CONNECTOR FOR CIRCUITS OF ANTIFERRORESONANCE RESISTOR													
	-XDA5				CONNECTOR FOR CIRCUITS OF NEUTRAL (RESIDUAL) CURRENT TRANSFORMER				-XDX		SUPPORT TERMINAL BLOCK													
	-XDB				CONNECTOR FOR ISOLATION OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK) AUXILIARY CIRCUITS					-XE1	EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, LEFT SIDE													
	-XDB1				CONNECTOR FOR AUXILIARY CIRCUITS OF CIRCUIT-BREAKER (OR UPPER VOLTAGE TRANSFORMER TRUCK)				-XE2		EARTHING TERMINAL BLOCKS FOR LV COMPARTMENT, RIGHT SIDE													
	-XDB2				CONNECTOR FOR ADDITIONAL AUXILIARY CIRCUITS OF CIRCUIT-BREAKER					-XE5	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, VOLTAGE INDICATOR SIDE													
	-XDB10...-XDB89				CONNECTOR FOR CIRCUIT-BREAKER INTERNAL CIRCUITS				-XE6		EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, REAR BOTTOM SIDE													
	-XDB7				CONNECTOR FOR TRUCK POSITION CONTACTS					-XE7	EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, LEFT SIDE													
	-XDB91				CONNECTOR FOR OPENING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR				-XE8		EARTHING TERMINAL BLOCKS FOR MV COMPARTMENT, FRONT BOTTOM SIDE													
	-XDB92				CONNECTOR FOR CLOSING RELEASE CIRCUITS OF SWITCH-DISCONNECTOR					-XE9	EARTHING TERMINAL BLOCK FOR TRANSFORMERS IN BUSBAR COMPARTMENT													
	-XDB93				CONNECTOR FOR POSITION SWITCHES OF SWITCH-DISCONNECTOR																			
	-XDB95				CONNECTOR FOR CIRCUITS OF EARTHING SWITCH LOCKING MAGNET																			
D	-XDC				CUSTOMER TERMINAL BLOCK																			
	-XDC1				CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF CIRCUIT BREAKER																			
	-XDC2				CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF SWITCH-DISCONNECTOR																			
	-XDC3				CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF PROTECTION RELAY AND MULTIFUNCTION UNIT																			
	-XDC4				CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF VOLTAGE TRANSFORMERS																			
	-XDC5				CUSTOMER TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF MINIATURE CIRCUIT BREAKER																			
	-XDE				TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE																			
	-XDE1				TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE																			
	-XDE2				TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2																			
	-XDE3				TERMINAL BLOCK FOR INTERNAL CIRCUITS OF EARTHING SWITCH -QCE1 OR -QCE2																			
	-XDF				TERMINAL BLOCK FOR AUXILIARY CIRCUITS OF FORCED COOLING VENTILATORS																			
	-XDF1				CONNECTOR FOR ISOLATION OF CIRCUITS OF FRONT VENTILATOR																			
	-XDF2				CONNECTOR FOR REAR VENTILATOR CIRCUITS																			
	-XDF3				CONNECTOR FOR FRONT VENTILATOR CIRCUITS																			
	-XDH				TERMINAL BLOCK FOR VARIOUS CIRCUITS (HEATERS, POSITION SWITCHES DETECTING INTERNAL ARCING, ETC.)																			
	-XDI				TERMINAL BLOCK FOR INTERCONNECTION (CONNECTION BETWEEN PANELS)																			
F					Date Prep.	15.11.2018	PROJECT TITLE		ABB		EPMV	REFERENCE DESIGNATIONS		Scale	NTS	= H02,H03								
					Prepared	YR	MV Switchgear for MNA Serang Plant			MVD-22 20kV Central WTP Area		Customer	PT Multimas Nabati Asahan	& 306										
	00	Issued for Approval	15.01.19	YR	Checked	AW			PT. ABB Sakti Industri		SCHEMATIC DIAGRAM		Drawing Nbr.	18.042-MNA-306.04	Sh. 6									
	Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042				SBC-W OUTGOING		Customer Doc. Nbr.		Cont. 27									
	1			2			3			4			5			6			7			8		

WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.



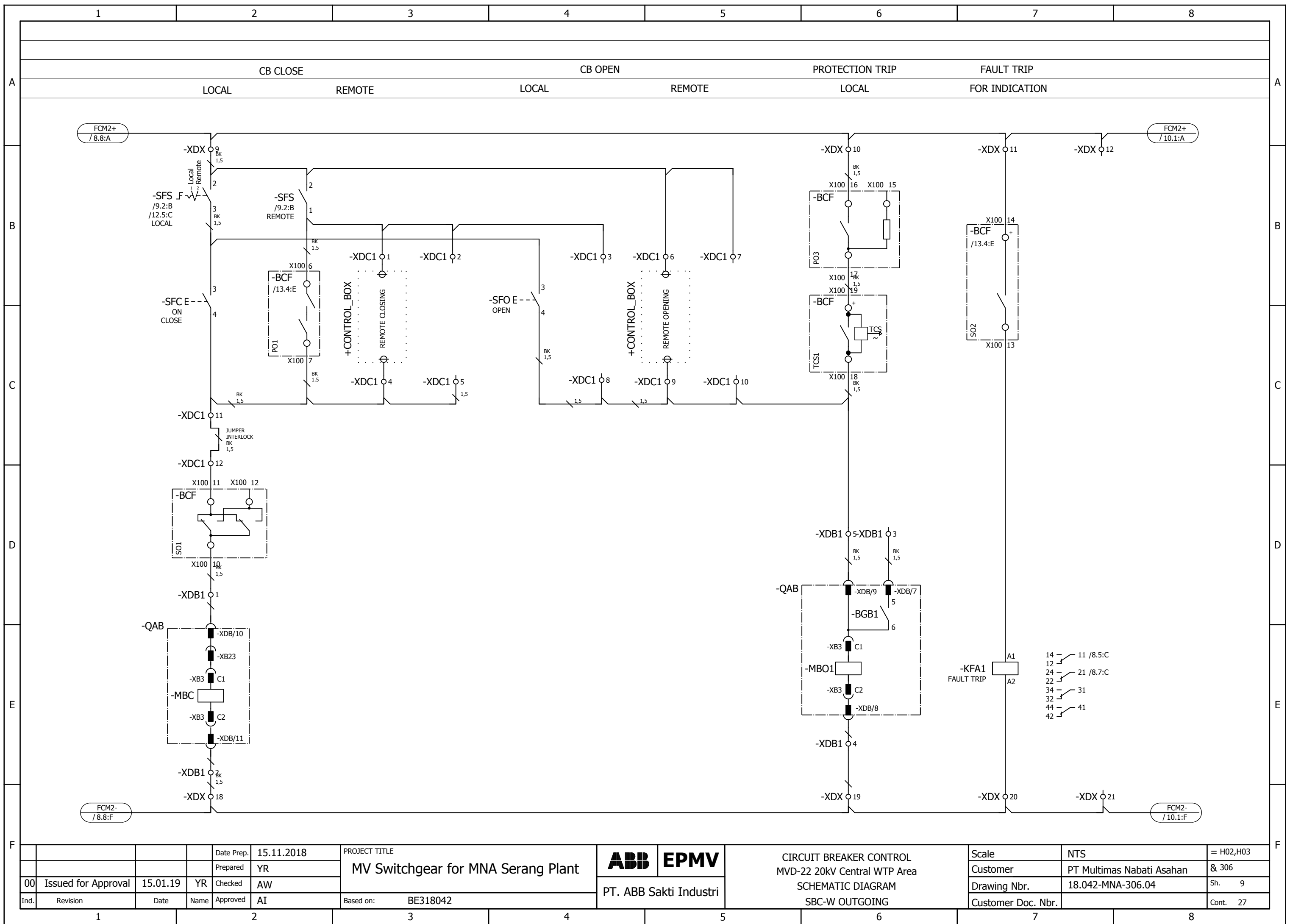
				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	DISTRIBUTION OF AUXILIARY CIRCUITS MVD-22 20kV Central WTP Area SCHEMATIC DIAGRAM SBC-W OUTGOING	Scale	NTS	= H02,H03
				Prepared	YR					Customer	PT Multimas Nabati Asahan	& 306
00	Issued for Approval	15.01.19	YR	Checked	AW		PT. ABB Sakti Industri	Drawing Nbr.		18.042-MNA-306.04	Sh. 7	
Ind.	Revision	Date	Name	Approved	AI			Based on: BE318042		Customer Doc. Nbr.		Cont. 27

WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.



				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant		ABB EPMV	AUXILIARY SUPPLY CIRCUIT MVD-22 20kV Central WTP Area SCHEMATIC DIAGRAM SBC-W OUTGOING			Scale	NTS	= H02,H03
				Prepared	YR							Customer	PT Multimas Nabati Asahan	& 306
00 Issued for Approval		15.01.19	YR	Checked	AW							Drawing Nbr.	18.042-MNA-306.04	Sh. 8
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042		PT. ABB Sakti Industri				Customer Doc. Nbr.		Cont. 27

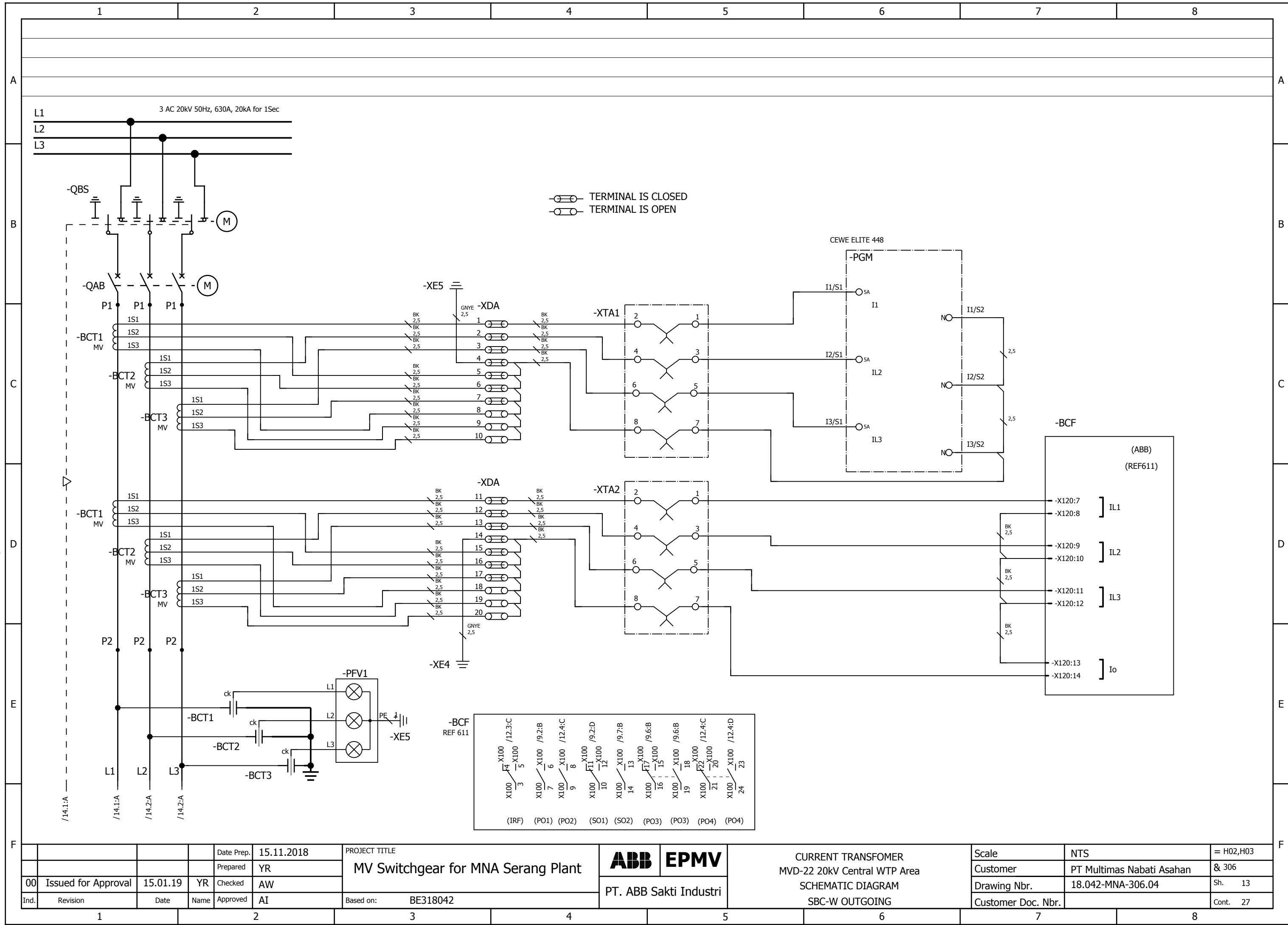
WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.



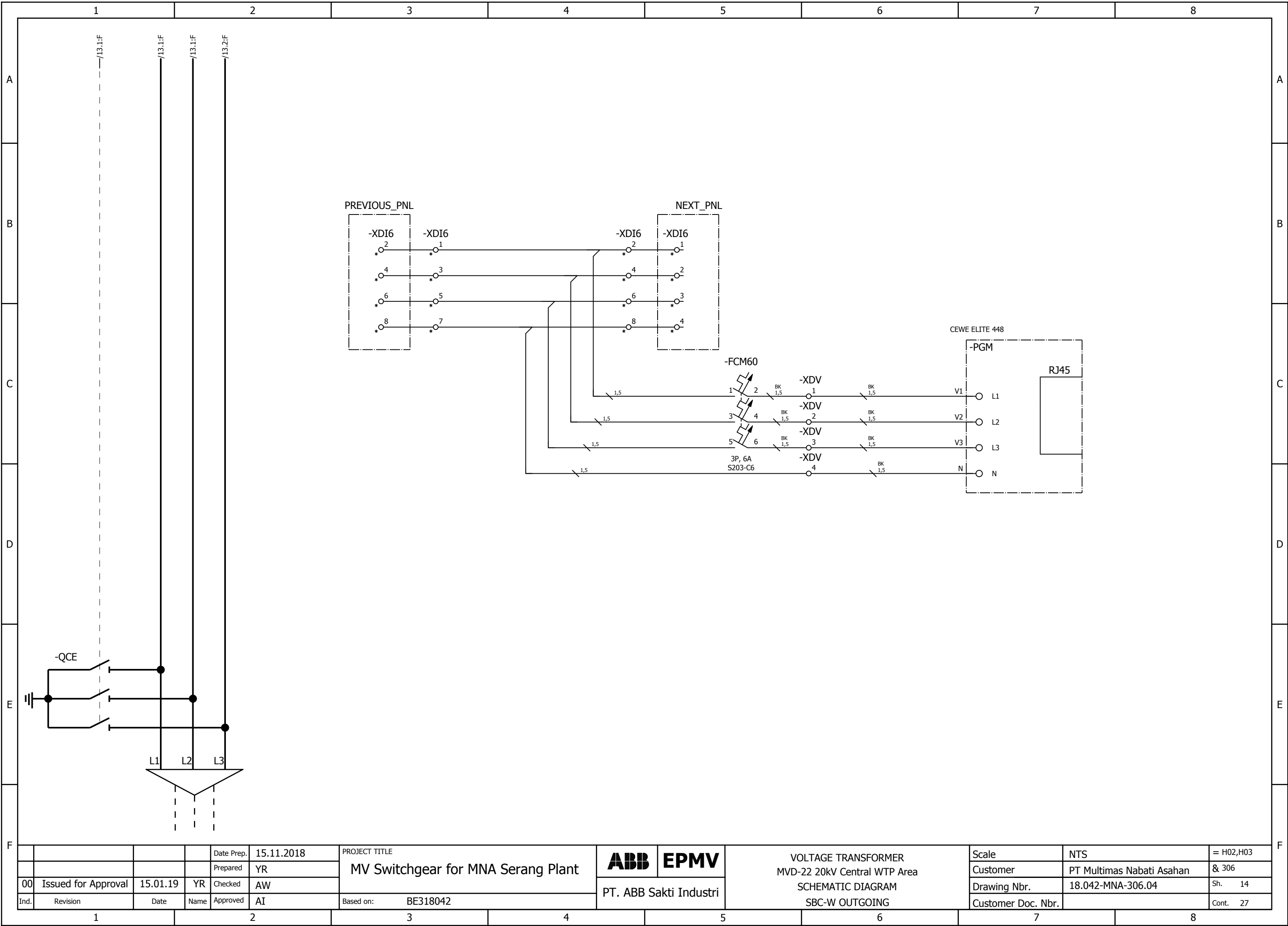
				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	AUXILIARY MOTOR & RELAY MVD-22 20kV Central WTP Area SCHEMATIC DIAGRAM SBC-W OUTGOING	Scale	NTS	= H02,H03
				Prepared	YR					Customer	PT Multimas Nabati Asahan	& 306
00	Issued for Approval	15.01.19	YR	Checked	AW		Drawing Nbr.	18.042-MNA-306.04		Sh. 10		
Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318042	Customer Doc. Nbr.			Cont. 27	
						PT. ABB Sakti Industri						

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	DIGITAL INPUT/OUTPUT RELAY MVD-22 20kV Central WTP Area SCHEMATIC DIAGRAM SBC-W OUTGOING	Scale	NTS	= H02,H03
				Prepared	YR					PT. ABB Sakti Industri	Customer	PT Multimas Nabati Asahan
00	Issued for Approval	15.01.19	YR	Checked	AW		Drawing Nbr.	18.042-MNA-306.04			Sh. 11	
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	Customer Doc. Nbr.			Cont. 27		

WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.

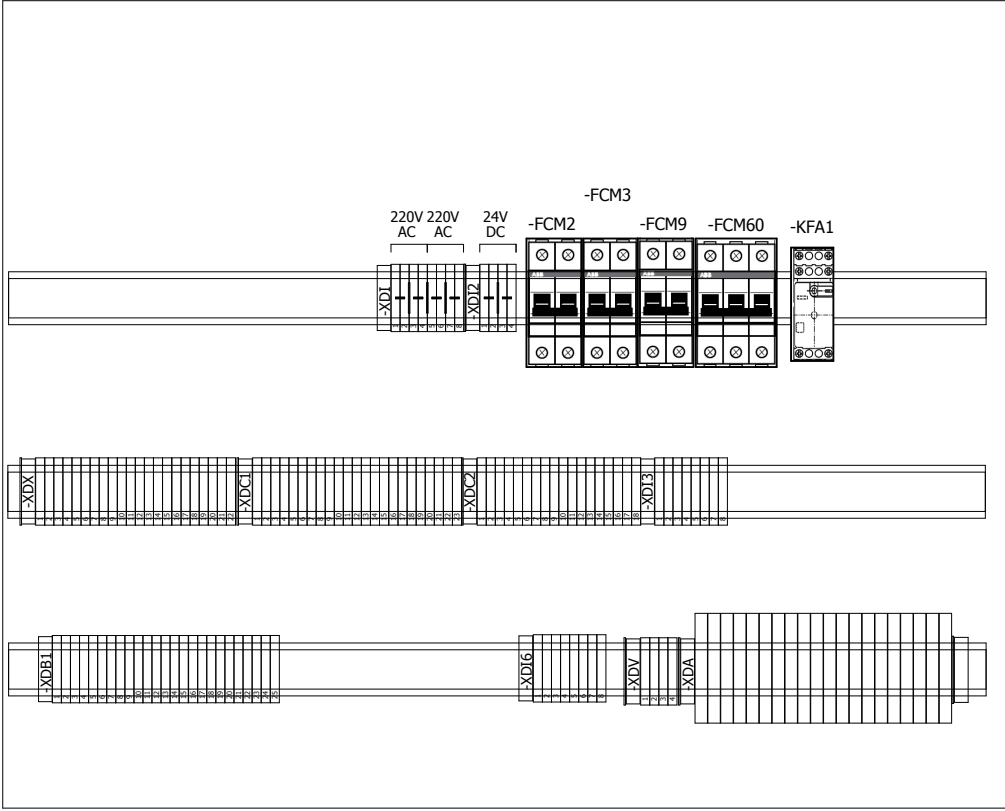


WE RESERVE ALL RIGHTS IN THIS DOCUMENT AND IN THE INFORMATION CONTAINED THEREIN. REPRODUCTION, USE OR DISCLOSURE TO THIRD PARTIES WITHOUT EXPRESS AUTHORITY IS STRICTLY FORBIDDEN.



				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	VOLTAGE TRANSFORMER MVD-22 20kV Central WTP Area SCHEMATIC DIAGRAM SBC-W OUTGOING	Scale	NTS	= H02,H03
				Prepared	YR					Customer	PT Multimas Nabati Asahan	& 306
00	Issued for Approval	15.01.19	YR	Checked	AW		Drawing Nbr.	18.042-MNA-306.04		Sh. 14		
Ind.	Revision	Date	Name	Approved	AI	Based on:	BE318042	PT. ABB Sakti Industri		Customer Doc. Nbr.		Cont. 27

No.	EQUIPMENT DESCRIPTION	ENGRAVING
1	-SFS (LOCAL-OFF-REMOTE SELECTOR SWITCH, K&N)	-SFS, LOCAL-OFF-REMOTE
2	-SFC (PUSH BUTTON CB CLOSE, GREEN, ABB)	-SFC, CB CLOSE
3	-SFO (PUSH BUTTON CB OPEN, RED, ABB)	-SFO, CB OPEN
4	-PFY1 (LAMP INDICATION FOR CB TRIP, YELLOW, ABB)	-PFY1, CB TRIP
5	-PFR1 (LAMP INDICATION FOR CB CLOSE, RED, ABB)	-PFR1, CB CLOSE
6	-PFG1 (LAMP INDICATION FOR CB OPEN, GREEN, ABB)	-PFG1, CB OPEN
7	-PGJ (DIGITAL MEASUREMENT, CEWE 448)	-PGJ, DIGITAL MEASUREMENT
8	-BCF (FEEDER PROTECTION RELAY, REF615J, ABB)	-BCF, RELAY PROTECTION
9	-XTA1, XTA2 (TEST BLOCK , YONGSUNG)	-XTA, CURRENT TEST BLOCK

[illegible]

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	LV BOX LAYOUT MVD-22 20kV Central WTP Area SCHEMATIC DIAGRAM SBC-W OUTGOING	Scale	NTS	= H02,H03
				Prepared	YR							Customer
00	Issued for Approval	15.01.19	YR	Checked	AW					Drawing Nbr.	18.042-MNA-306.04	Sh. 16
Ind.	Revision	Date	Name	Approved	AI	Based on: BE318042	PT. ABB Sakti Industri			Customer Doc. Nbr.		Cont. 27

Carry over: 0.00

[illegible]

				Date Prep.	15.11.2018	PROJECT TITLE MV Switchgear for MNA Serang Plant	ABB	EPMV	PART LIST MVD-22 20kV Central WTP Area SCHEMATIC DIAGRAM SBC-W OUTGOING	Scale	NTS	= H02,H03
				Prepared	YR					PT. ABB Sakti Industri	Customer	PT Multimas Nabati Asahan
00	Issued for Approval	15.01.19	YR	Checked	AW		Drawing Nbr.	18.042-MNA-306.04			Sh. 17	
Ind.	Revision	Date	Name	Approved	AI		Based on: BE318042	Customer Doc. Nbr.			Cont. 27	

[illegible]

F										E										D										C										B										A									

F		E		D		C		B		A	
</											

[illegible]

F										E										D										C										B										A																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																	
Ind. RevisionDateNameApproved										REMARK										CABLE										EQUIPMENT										WIRE NO.										TYPE										Terminal number										Wire No										EQUIPMENT										CABLE										REMARK										CABLE										EQUIPMENT										WIRE NO.										TYPE										Terminal number										Wire No										EQUIPMENT										CABLE										REMARK																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																					
										BK 2,5 mm²										-FCM9:1																														PT 2.5										1																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																																							

[illegible]

[illegible]

[illegible]

[illegible]

F										E										D										C										B										A									